

Telecom Customer Churn Prediction Using Machine Learning

PROBLEM STATEMENT

Goal: Predict customers likely to leave before they churn.

Q: What is customer churn in telecom industry?

A: Customer churn occurs when customers stop using a telecom service and switch to another provider.

Business Problem

- Acquiring new customers is expensive
- Losing existing customers reduces revenue
- Telecom companies need early identification of at-risk customers

PROJECT OBJECTIVES

01

Analyze telecom customer behavior.

04

Compare model performances.

02

Identify factors contributing to churn.

05

Generate actionable retention strategies.

03

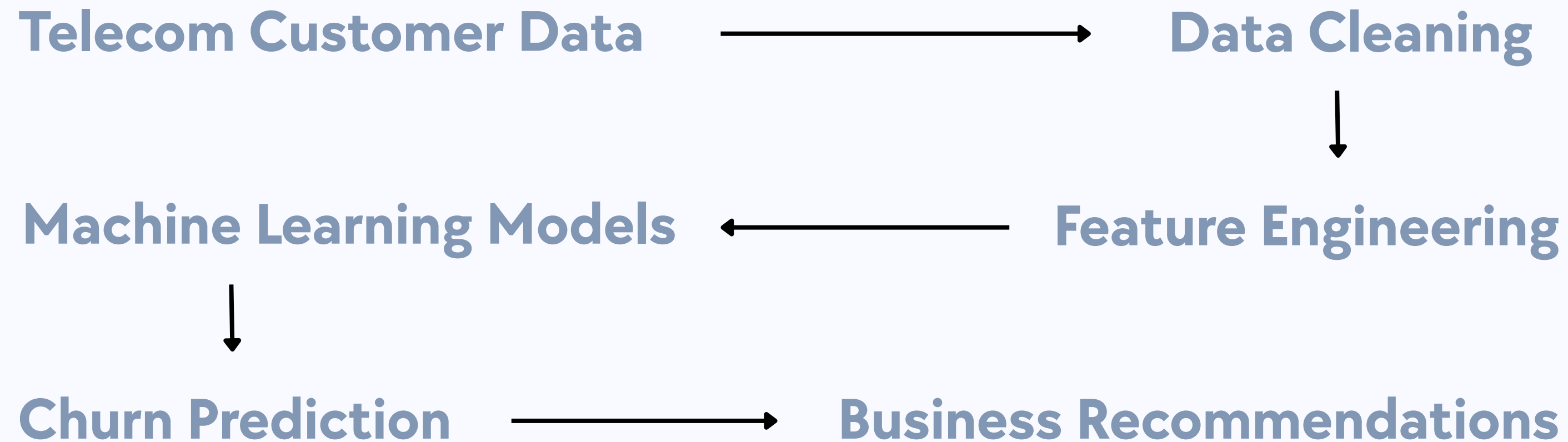
Build predictive machine learning models.

06

Evaluate model performance using classification metrics.

Dataset & Methodology

Project Workflow

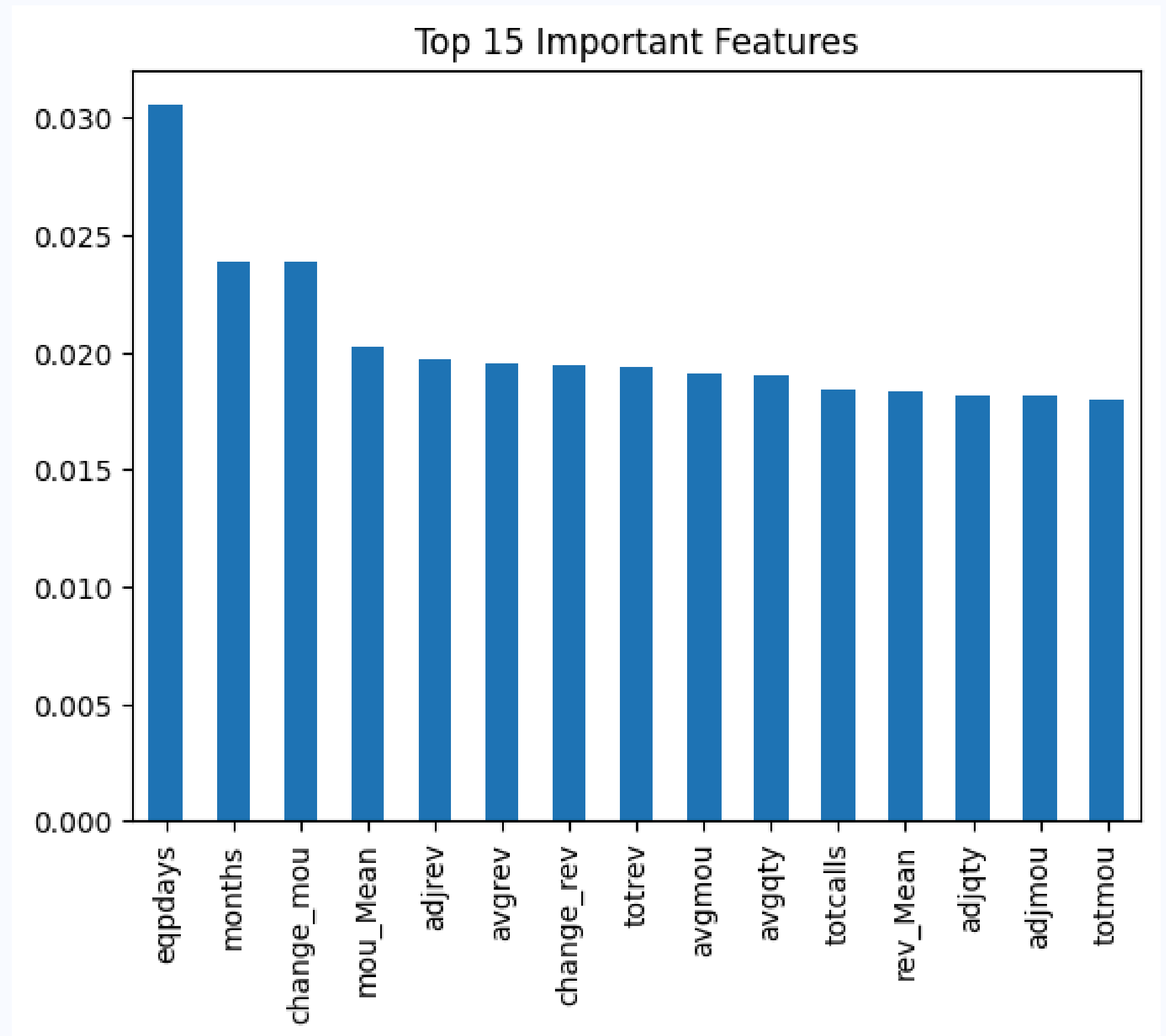


FEATURE IMPORTANCE

Q: What factors influence customer churn?

A: Analysis of the feature importance results indicates that:

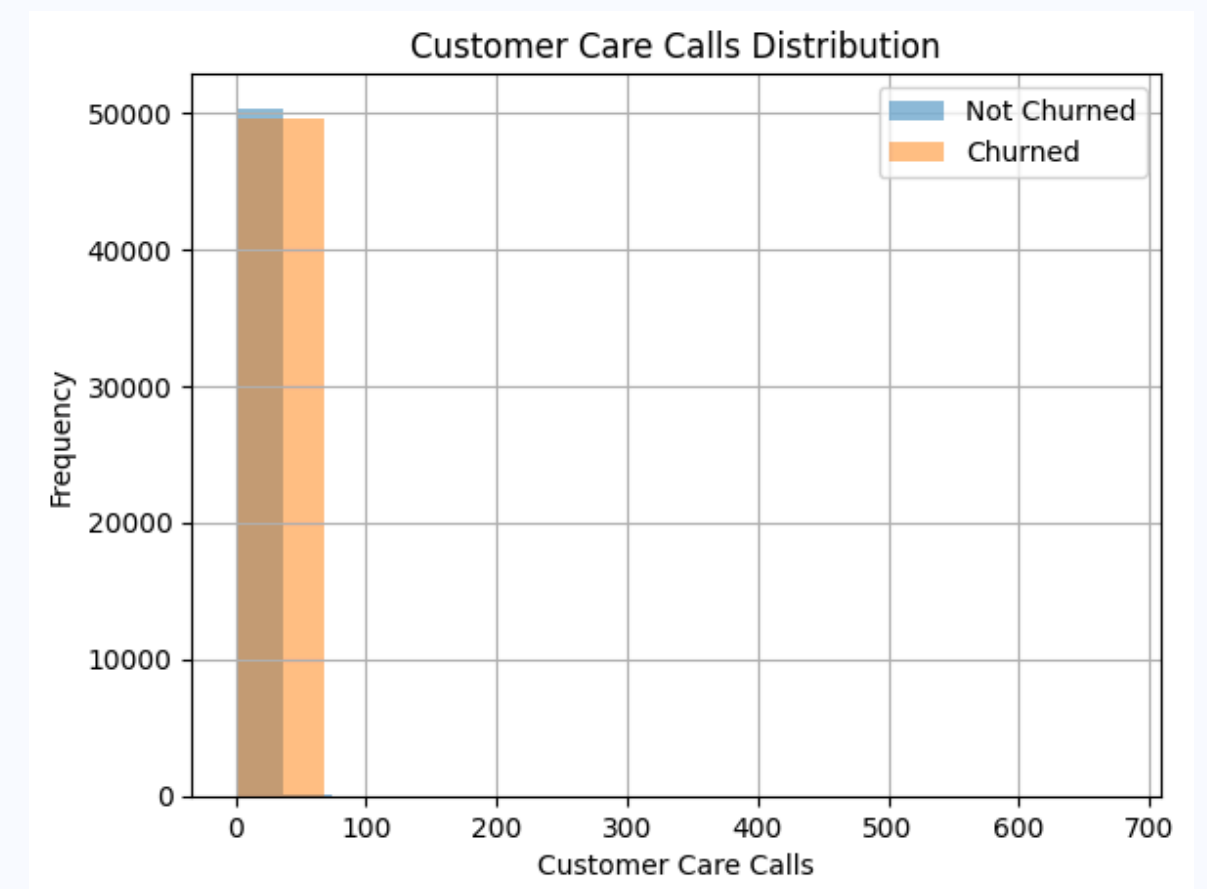
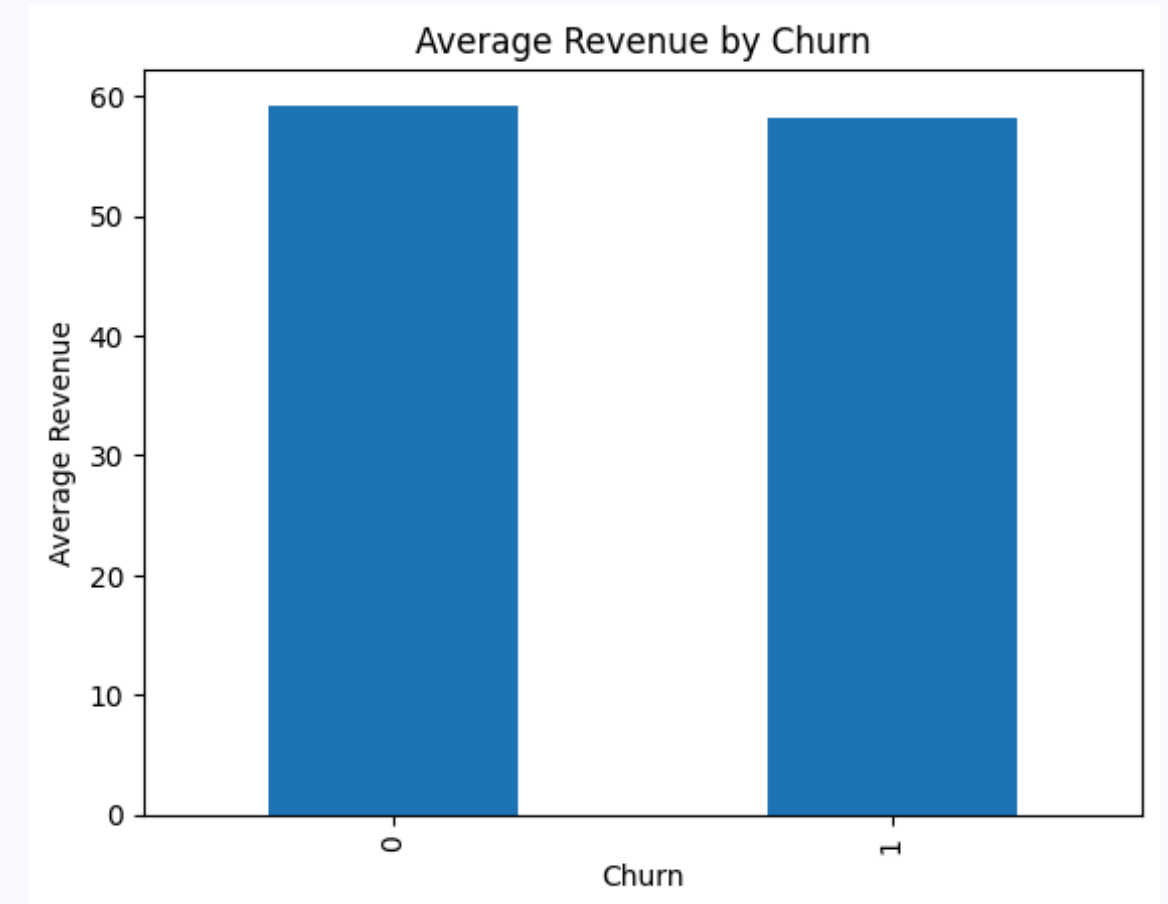
- Service quality metrics strongly affect churn.
- Customer support interactions correlate with churn.
- Customer tenure influences retention.
- Usage patterns provide predictive information.



CUSTOMER BEHAVIOR ANALYSIS

Q: What distinguishes churning customers from loyal customers?

Churning Customers	Loyal Customers
Shorter service duration	Longer service duration
More customer care interactions	Fewer customer support requests
Higher service dissatisfaction	Better service experience
Lower customer retention	Higher customer retention



ROLE OF MACHINE LEARNING

User Information
(Service Quality,
Usage, Revenue,
Customer Care)



ML Models (Logistic
Regression, Random
Forest, XGBoost)



High / Low Risk



Probability of Churn

MODEL PERFORMANCE

Q: Which model performs best?

A: XGBoost achieved the highest accuracy (64.1%) and was selected as the final model for churn prediction.

Model	Approach / Purpose	Accuracy
Logistic Regression	Baseline linear classification model used as a benchmark	59.5%
Random Forest	Ensemble model using multiple decision trees to improve prediction robustness	61.8%
XGBoost	Gradient boosting ensemble model designed for high predictive performance	64.1%

BUSINESS IMPACT

Customer Data



Predict Churn



Identify High-Risk Customers



Targeted Retention Campaigns



Reduced Customer Loss

Benefits:

- Better allocation of marketing resources
- Enhanced customer satisfaction
- Improved customer retention
- Reduced revenue loss

RETENTION STRATEGIES

Recommendation	Proposed Action
Improve Service Quality	Expand network coverage in underserved and remote regions, reduce dropped calls, and improve connection reliability.
Strengthen Customer Support	Increase availability of human support agents while continuously improving AI-assisted customer service for faster issue resolution.
Target High-Risk Customers	Use churn prediction scores to provide personalized discounts, bundled plans, partner subscription offers (e.g., streaming or entertainment services), and retention incentives to at-risk customers.
Customer Loyalty Programs	Reward long-term customers with tiered benefits such as discounts, priority support, exclusive offers, and loyalty rewards based on years of service.

KEY TAKEAWAYS

- ✓ Customer churn can be predicted using ML.
- ✓ Service quality significantly affects churn.
- ✓ XGBoost achieved the best performance.
- ✓ Predictive analytics can support retention strategies.